Parity barrier in primes over function field zeta

A dimensional obstruction, measured

First edition. Packaged from the twin-prime-parity programme, phases 018, 019 and 020. The manuscript reports what the programme established, what it measured, and what it did not prove. It is a negative-results paper with a positive core: the obstruction is not an estimate to be sharpened but a dimensional invariant, and in the polynomial model it can be measured exactly.

Abstract

Over `F_q[x]`, "primes" are monic irreducibles (qrimes) and "twin primes" are pairs `(f, f+2)`. The twin count `T(n)` is a bilinear statistic, while the zeta function of the ring determines only linear (degree-count) data; this separation is proved with an explicit witness. The programme's central question is constructive: can a sieve, a degree window, or a second-moment argument cross the parity barrier in this model?

We report four measured failure modes. The Selberg `Λ²` upper bound at the classical level `n/2` carries the classical Selberg constant `2^κ κ!` — `8×` the truth for the twin problem (`κ = 2`) and `2×` for the qrime problem (`κ = 1`). Over `n ≤ 14` the twin ratio hovers near 4, which an earlier version of this paper mistook for the limit; computed from the same series it reaches `7.91` at `n = 1000`. The correction is marked below. The constant is shift-independent: a polynomial shift divisible by every linear prime has the same limit 8 against its own singular series. The bilinear pair weight is indefinite and defines no bound at all. Degree windows give no `L^{−1/2}` cancellation. The large sieve is slack by a factor of order `n`. The exact coprime-pair expansion of `T(n)` and the degree grading that makes it exact are both machine-checked in Lean 4.

On the quantitative side, `T(n) = S(q)q^n/n² + E(n)` holds with `|E|/q^{n/2} ≤ 0.30` for every `n ≤ 14` at `q = 3` (`T(13) = 3747`, `T(14) = 9510`), and the corresponding character-sum identity of the programme is corrected at 80 primes: the original form holds exactly when `q ≡ 1 (mod 8)`, the corrected form for every `q ≡ 1 (mod 4)` tested. None of this proves the fixed-q asymptotic; that statement is conjectural in the literature we surveyed.

Conclusion. The barrier is a missing *object* — a way to resolve the two conditions jointly — not a missing estimate. Over `F_q[x]` that object is the Frobenius; over $\mathbb{Z}$ it would be a grading substitute for the Möbius truncation. Four standard mechanisms are now closed by measurement, and every proof strategy must reproduce the numbers recorded here — including the corrected constant 8, not 4.

Correction notice

This edition corrects two claims, and marks both as corrected.

C0 — the "dimension law" (this paper's own previous version)

An earlier version of this paper stated that the Selberg `Λ²` bound at level `n/2` is `2^dim` times the truth asymptotically, `4×` for twins and `2×` for qrimes, and called 4 "the method's limit". That is false. Under the paper's own definition the ratio tends to the classical Selberg constant `2^κ κ!`: 8 for twins (`κ = 2`), 2 for qrimes (`κ = 1`).

The finite table was an exact optimum at each `n`, not the value of the limit: the twin ratio is `3.61, 3.76, 4.01, 4.00` at `n = 6, 8, 10, 12`, and then `5.12, 6.26, 7.02, 7.48, 7.72, 7.86, 7.91` at `n = 20, 40, 80, 160, 300, 600, 1000`. The value near 4 at `n ≈ 12` is a pre-asymptotic transient.

The correction was found by the project's referee (Benjaminsen, review 1736), whose independent computation this edition reproduces from the same series; section 16 records the exchange and what changed. The qualitative conclusion is unchanged and slightly stronger: the sieve loses a factor 8, not 4.

C1 — the character-sum identity of phase 019

The identity was recorded as

$\Sigma_A \chi_2(A)\,|K(A,c)|^2 = 2\,\chi_2(c)\,\sqrt{q}\,\mathrm{Re}\,J(\chi_4,\chi_4)^2$ (as stated)

That statement is false at its stated scope. Tested exactly in $\mathbb{Z}[\zeta_q]$ at every prime $q \equiv 1 \pmod 4$ up to 1000, it holds at all 37 primes $q \equiv 1 \pmod 8$ and fails at all 43 primes $q \equiv 5 \pmod 8$. The corrected statement is

$\Sigma_A \chi_2(A)\,|K(A,c)|^2 = 2\,\chi_2(2c)\,\sqrt{q}\,\mathrm{Re}\,J(\chi_4,\chi_4)^2$ (corrected)

which holds at all 80 primes tested. The one-character factor $\chi_2(2c)$ is the entire correction, and the hypothesis is exactly $q \equiv 1 \pmod 8$. The Lean-certified combinatorial step (`Twist.twisted_parseval_step`) is unaffected — what changes is the arithmetic evaluation, not the splitting.

Four further corrections carried by this edition (the invalid truncation, the ω counter, an over-stated completeness claim, and a non-existent closed form) are listed in §13, each beside the claim it corrects.

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1. Introduction

1.1 The transposition

The twin prime conjecture asks whether n and $n+2$ are both prime infinitely often. Over $\mathbb{F}_q[x]$ the analogue has a known answer in a large-q regime (Pollack; Bary-Soroker–Stix) and is conjectural at fixed q: the large-sieve error $O(q^{n-1/2})$ is useless as $n \to \infty$. The model is therefore a genuine testbed: the statements are precise, the computations are exact, and the obstruction is the same parity phenomenon that blocks $\mathbb{Z}$.

The programme asks a narrower question than "prove the conjecture": which statistics can each tool see, and where exactly does each standard mechanism fail? The answer is the content of this paper.

1.2 What is new here

1. A measured constant: the square-weight sieve deficiency is the classical Selberg constant $2^\kappa\,\kappa!$ (8 for twins), computed exactly in $\mathbb{F}_q[x]$ and confirmed to $n = 1000$; it is invariant under polynomial shifts that absorb every small prime (§3.3, §4.1, §7). An earlier version stated $2^{\dim} = 4$; correction C0 records why that was wrong.
2. An exact expansion of the twin count with a boundary stratum whose coefficients are 0 or 1, both machine-checked, together with the fact that the natural degree truncation diverges while the boundary cancels it (§5).
3. A closed set of failure modes for the four standard mechanisms, each with a number (§4).
4. An exact test of the programme's residual character-sum identity, which is false as stated and true after a one-character correction, with the precise hypothesis $q \equiv 1 \pmod 8$ (§6).
5. Numerical support for the fixed-q asymptotic to $n = 14$, with the error always at the Weil scale (§8).

Nothing here proves the twin prime conjecture, the parity problem over $\mathbb{Z}$, or the Riemann hypothesis.

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2. The model

Fix an odd prime power q. Let $I(n)$ be the number of monic irreducibles of degree n and

$$T(n;h) = \#\{f \text{ monic}, \deg f = n : f \text{ and } f+h \text{ both irreducible}\}, \qquad T(n) := T(n;2).$$

The singular series is

$$S(q) = \prod_P \left(1 - \frac{2}{N(P)}\right) \Big/ \left(1 - \frac{1}{N(P)}\right)^2,$$

the product over monic irreducibles P, with $N(P) = q^{\deg P}$. It converges absolutely. The programme's main term is $S(q)\,q^n / n^2$, the function-field Hardy–Littlewood prediction.

The zeta of a set of qrimes. For a set S of monic irreducibles put $Z(S)(n) = \#\{f \in S : \deg f = n\}$ and $\zeta_S(T) = \prod_{f \in S} (1 - T^{\deg f})^{-1}$. Then $[T^n] \log \zeta_S = (1/n) \sum_{d \mid n} d\, Z(S)(d)$, so ζ_S and the degree-count sequence $Z(S)$ determine each other exactly (Euler product and Möbius inversion). This is the "zeta datum" the separation theorem is about: the twin count is a bilinear statistic of the set, and the zeta records only the linear one.

Two elementary facts are used throughout. First, the degree grading: every nontrivial factorisation of a degree-n polynomial has a factor of degree ≤ n/2 (Lean: `Grading.exists_two_mul_le`). Second, the local dichotomy: at a prime of norm Q, avoiding 0 leaves Q−1 residues, while avoiding both 0 and −h leaves Q−2 (Lean: `Capstone.local_factor_dichotomy`).

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3. The barrier, in three statements

3.1 The zeta cannot see the pair

Theorem (separation). The zeta function of a set of qrimes determines exactly its degree counts, and no function of those counts determines the twin count.

The proof exhibits two sets of four degree-2 qrimes over $\mathbb{F}_7$ with identical degree profiles and twin counts 1 and 0. The general statement is machine-checked (`Separation.no_zeta_observable`); the witness is `Separation.separation_F7`.

The consequence is structural: the twin count is *bilinear* data, and the zeta is *linear* data. No amount of zeta computation recovers it.

3.2 Bounded-support divisor sums cannot see it

Theorem (parity). Every divisor sum with bounded support is periodic; the twin indicator is not. Hence no finite-level Selberg sieve weight equals the twin indicator.

This is the finite form of Selberg's parity principle. The two halves are Lean-checked (`Periodic`, `NonPeriodic`). The theorem concerns the *indicator*, not the count: it excludes a class of weights, and therefore any route must use growing support.

3.3 Growing support loses the classical Selberg constant

The one route the parity theorem leaves open is a growing-support sieve. Its best square-weight version is computed exactly, and it loses a constant:

Measured (Selberg constant). With `M[d,e] = 2^{ω(lcm(d,e))} q^{−deg lcm(d,e)}` and `bound = q^n/(M^{−1})_{1,1}`, put `F(u) = Π_P (1 + 2u^{deg P}/(q^{deg P} − 2))`. Then $F(u) = H(u)/(1-u)^2$ with H analytic at u = 1 and `H(1) = 1/S(q)`, so with `m = n/2`

$$G(m) = [u^m]\, F(u)/(1-u) \sim m^2/(2\,S(q)), \qquad \text{bound} \sim 2\,S(q)\,q^n/m^2 = 8\cdot S(q) q^n/n^2 .$$

The ratio therefore tends to `2^κ κ!` — 8 for twins, 2 for qrimes. This is the classical Selberg-sieve constant (Halberstam–Richert, *Sieve Methods*, ch. 1; Iwaniec–Kowalski, ch. 6–7), not a new law: the `F_q[x]` computation reproduces it exactly and confirms it numerically to `n = 1000`.

Two consequences are load-bearing. First, the constant is not `2^dim`: the value 4 seen at `n ≤ 12` is a transient (correction C0). Second, the constant is shift-independent (§7): the same 8 appears for a polynomial shift that divides every small prime. That is what the phase's measurements actually establish: the deficiency is a *dimension-dependent constant*, not a small-n artefact of the local densities.

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4. The four mechanisms, measured

All measurements are exact finite computations; the audit script re-checks 35 of them.

4.1 The square-weight sieve

With `M[d,e] = 2^{ω(lcm(d,e))} q^{−deg lcm(d,e)}` and `bound = q^n/(M^{−1})_{1,1}`:

q	n	bound	T(n)	bound / T(n)
3	6	21.67	6	3.61
3	8	135.37	36	3.76

q	n	bound	T(n)	bound / T(n)
3	10	866.41	216	4.01
3	12	6020.30	1506	4.00
5	6	966.94	240	4.03

These rows are exact, and they are not the limit. The matrix is infeasible past `n ≈ 20`, but the equivalent Selberg series `G(m)` is not, and it gives the same numbers (to three decimals) and then keeps going:

n	12	20	40	80	160	300	600	1000
bound / main	4.12	5.12	6.26	7.02	7.48	7.72	7.86	7.91

consistent with `8(1 − c/m)`. The value `≈ 4` at `n ≈ 12` is a transient: the ratio is monotonically rising. The dimension-1 control (counting qrimes) behaves the same way toward 2: `1.10` at `n = 4`, `1.57` at 12, `1.85` at 40, `1.98` at 300.

4.2 The bilinear pair weight

The natural bilinear weight gives the matrix `C[d,e] = #{f : d | f, e | f+2}`. Its eigenvalues at `q = 3`, `κ = 1` are `(−27, −27, −20.65, 317.65)` at `n = 5`, and negative eigenvalues persist at every `n` tested. The form is indefinite, so there is no minimum and no bound. The square in `Λ²` is not cosmetic: it is what makes the weight nonnegative.

4.3 Degree windows

Windowed averages of the main term do not cancel like `L^{−1/2}`. At `q = 3`, with the relative error measured against `S(q)q^n/n²`: pointwise errors fall from `25.2%` at `n = 6` to `3.0%` at `n = 12`; length-4 windows range from `0.34%` (over `n = 6..9`) to `4.23%` (over `n = 7..10`). The improvement is position-dependent, not length-dependent — the error is a deterministic oscillation, not noise, so a window is not a self-validating estimator.

4.4 The large sieve

For the twin indicator, `L(κ)/R(κ)` at level `κ = n/2` is `0.161, 0.069, 0.077, 0.056, 0.076, 0.045` for `n = 5..10`, decaying roughly like `1/n`. The bound extracted is `≍ q^n/n`, larger than the main term by a factor `n`. This is the concrete form of the known statement that the large-sieve error `O(q^{n−1/2})` is useless at fixed `q`.

4.5 Summary

mechanism	failure	factor
`Λ²` square weight at level `n/2`	capped	`2^κ κ!` (8 for twins, 2 for qrimes)
bilinear pair weight	indefinite	undefined
degree windows	no cancellation	position-dependent
large sieve	slack in the second moment	`≍ n`

The four failures have one cause: none resolves the two local conditions jointly and exactly.

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5. The exact side

5.1 The expansion

For `f` of degree `n`, the Legendre sieve over primes of degree `≤ n/2` is an exact indicator (`1_Irr`), by the degree-grading lemma. Expanding the product gives an exact, finite sum over coprime divisor pairs:

```
T(n) = Σ_{d,e | P} μ(d) μ(e) · #{f : deg f = n, d | f, e | f+2},
```

where P is the product of the primes of degree ≤ n/2. The CRT count is q^{n−deg d−deg e} when deg d + deg e ≤ n, and 0 or 1 on the boundary deg d + deg e > n — the boundary coefficients are 0/1 because a residue class modulo m with deg m > n contains at most one polynomial of degree n (Lean: Boundary.unique_of_deg_lt).

The expansion reproduces the brute-force count exactly (8 test cases), and it exposed a real defect in an earlier draft of the programme: dropping the boundary terms is invalid.

5.2 Two truncations, not one

Truncating the expansion at k = deg d + deg e ≤ n gives a partial product of a divergent product: its logarithm is ≈ −2 Σ_{d ≤ n/2} 1/d → −∞. The truncation is off by factors 10−100 and changes sign. The Euler product truncated by prime degree (deg Q ≤ n/2) instead tracks T(n) to about 10%, but y²P(y) has no limit (0.070 → 0.115 for y ≤ 12), so it is an approximation, not a closed form.

This distinction matters: "the truncated product diverges" is true of the degree truncation only.

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6. The character-sum identity of 019

Phase 019 tests the programme's residual identity

Σ_A χ₂(A) |K(A,c)|² = 2 χ₂(c) √q Re J(χ₄,χ₄)²,

with K(A,c) = Σ_w χ₂(A² − w²) ζ^{−cw/4}. The test is exact: both sides are elements of $\mathbb{Z}[\zeta_q]$, compared modulo the cyclotomic polynomial. Results at every prime q ≡ 1 (mod 4) up to 1000:

- the identity as stated holds at all 37 primes ≡ 1 (mod 8) and fails at all 43 primes ≡ 5 (mod 8);
- the corrected form Σ_A χ₂(A)|K(A,c)|² = 2 χ₂(2c) √q Re J² holds at all 80;
- the untwisted identity Σ_A |K(A,c)|² = q² − 2q holds at all 80.

The corrected form holds at every q ≡ 1 (mod 4) tested; the original form is valid exactly when q ≡ 1 (mod 8) (because χ₂(2) = −1 precisely then). The corrected statement is the one a proof should target. This is the correction marked in the notice above: it is a correction to the programme's own earlier statement, not to the literature, and the corrected form is the one this paper carries. The combinatorial step "sum over squares = sum over all plus the χ₂-twist" is Lean-certified (Twist.twisted_parseval_step).

The identity itself remains open.

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7. Polynomial shifts

Constant shifts never share a prime with h, which is why the singular series is shift-free. For a polynomial shift h ∈ F_q[x] of positive degree, primes can divide h; at such a prime the two conditions coincide and the local factor becomes 1/(1−1/Q) instead of (1−2/Q)/(1−1/Q)². The ratio is (Q−1)/(Q−2), giving

S_h(q) = S(q) · Π_{Q | h} (Q−1)/(Q−2).

At q = 3, h = x gives S_h = 0.792208 = 2 S(3), and the measured ratios T(n;x)/(S_h q^n/n²) are 1.039, 0.997, 0.962, 0.985, 0.987 for n = 5..9. Polynomial shifts therefore obey the same asymptotic with the modified series.

Crucially, the constant is unchanged. For h = x(x+1)(x+2), divisible by every linear prime, the ratio rises through 1.94, 2.37, 2.42, 3.12, 2.83, 3.51, 3.21, 3.84 for n = 6..13, and the same Selberg series at larger n gives 5.73 (40), 7.30 (160), 7.61 (300): it approaches 8, not 2 — the same constant as the constant-shift problem, measured against its own S_h. The deficiency is dimensional and shift-independent; the earlier claim that it "approaches 4" was the same small-n transient as C0.

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8. The main term and the error

With `S(3) = 0.396104`, `S(5) = 0.709544`, `S(7) = 0.812706`:

n	T(n) (q = 3)	S(3)q^n/n²	E(n)	q^{n/2}	|E|/q^{n/2}
10	216	233.90	−17.90	243.0	0.074
11	585	579.91	+5.09	420.9	0.012
12	1506	1461.85	+44.15	729.0	0.061
13	3747	3736.79	+10.21	1262.7	0.008
14	9510	9666.09	−156.09	2187.0	0.071

The ratio stays in `[0.008, 0.30]` over `n = 2..14`; at `q = 5`, `T(9) = 17265` against `17108.99` predicted (ratio `0.112`). The main term is accurate to `0.3%` at `n = 13`.

The error is not spectral: `T(n)` satisfies no finite linear recurrence (Berlekamp–Massey on held-out terms fails for `q = 3, 5, 7`). The natural reading is an infinite `q^{n/2}` zero spectrum.

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9. Proved, measured, open

Proved (Lean 4, standard axioms; 61 modules in `018/lean`, `019/lean`, `020/lean`): the separation witness; the parity theorem; the degree grading; the boundary `0/1` lemma; the local factor `2^k` and the local-factor dichotomy; the shift law's two halves; the window algebra; the `(Q−1)/(Q−2)` correction; the 019 splitting step.

Measured (exact finite computation, 35 audited claims + the Selberg series to `n = 1000`): the constant `2^κ κ!` (8 for twins, 2 for qrimes); the error band; the orbit counts; the large-sieve slack; the polynomial-shift series; `T(13)`, `T(14)`.

Open: the fixed-`q` asymptotic; its polynomial-shift version; the 019 identity; the generic case of the `d=4` cap of phase 018.

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10. The transfer demand

The function-field crossing has two ingredients: an exact growing-support identity, and a canonical truncation of it. Over `F_q[x]` the degree supplies both. Over $\mathbb{Z}$ the analogous identity runs through `Λ = μ ∗ log`, whose support has no canonical truncation, and the truncation error *is* the two-point Chowla correlation.

So the demand is a grading substitute: a canonical, error-controlled truncation of the Möbius sum. Every standard $\mathbb{Z}$ method that supplies one — the circle method, the large sieve — is blocked at exactly this point, and §4 says why: their truncations lose the Selberg constant 8, a factor `n`, or worse.

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11. Limitations

1. Nothing here crosses the barrier. The paper measures it and closes four mechanisms. An earlier version stated the wrong constant — it claimed `2^dim = 4`; the correct limit is `2^κ κ! = 8` for twins. Correction C0 and §16 record the error, the referee who found it, and the fix.
2. The fixed-`q` asymptotic is not proved. The data to `n = 14` constrains it; it does not establish it.
3. The character-sum identity is verified, not proved.
4. The `d=4` cap of phase 018 depends on three cited theorems and one unformalised decomposition.
5. "Measured" means exact finite computation over the stated ranges. It is evidence of the stated strength, nothing more.

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12. Reproduction

```
./check.sh                      # all 61 Lean modules (needs Lean 4 + Mathlib at $HOME/lean)
python3 020/checks/audit.py     # 35 numeric claims
python3 -m unittest discover -s 020/tests
python3 -m unittest discover -s 019/tests
python3 020/checks/linkcheck.py
```

Python conventions: results go to stdout, diagnostics and banners to stderr; no script writes files. Recorded inputs live in `020/data/`.

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13. Corrections carried by this edition

Each correction is kept beside the claim it corrects; the first is the one marked in the notice above.

#	claim as recorded	correction	evidence
C0	the Λ^2 bound at level n/2 is 2^dim times truth: 4 for twins, 2 for qrimes	false: the ratio tends to 2^κ κ!, i.e. 8 for twins and 2 for qrimes; 4 was a transient at n ≈ 12	referee review 1736; §4.1 series to n = 1000; §3.3
C1	$\Sigma_A\ \chi_2(A)\lvert K(A,c)\rvert^2 = 2\ \chi_2(c)\ \sqrt{q}\ \mathrm{Re}\ J^2$	false at **q ≡ 5 (mod 8)**; corrected to 2 χ₂(2c), hypothesis q ≡ 1 (mod 8)	80 primes, exact $\mathbb{Z}[\zeta_q]$ arithmetic
C2	the coprime-pair expansion may be truncated at deg d + deg e ≤ n	invalid: the boundary terms exist and are 0/1, not zero	020/notes/round108.md; Lean Bound-ary.unique_of_deg_lt
C3	the constant c(κ) of round 109	over-counted repeated prime factors (only one copy of ω divided out)	020/notes/round109.md erratum; 0 mismatches after the fix
C4	"the sieve is complete for the entire Selberg class"	over-stated: complete for bounded-support weights only	020/notes/round118.md
C5	a closed form exists for the boundary stratum	no constant exists: $y^2P(y)$ has no limit (0.070 → 0.115)	020/notes/round119.md

14. Claim sources

Each numbered claim of this paper names where its evidence lives in the attached tree.

claim	source
separation theorem, F_7 witness	018/lean/Separation.lean; 018/checks/round3.py
parity theorem (periodic vs not)	018/lean/Periodic.lean, 018/lean/NonPeriodic.lean
degree grading ≤ n/2	020/lean/Grading.lean
boundary coefficients 0/1	020/lean/Boundary.lean
local factor 2^k, dichotomy	020/lean/Dimension.lean, 020/lean/Capstone.lean
Selberg constant 2^κ κ! (8 twins, 2 qrimes)	this edition's submission/selberg_constant.py; §4.1; referee review 1736
§4.1 finite table (21.67, 135.37, 866.41, 6020.30, 966.94)	020/checks/audit.py; 020/notes/round110.md; reproduced by the referee

claim	source
§4.1 large-n series (5.12 … 7.91)	`submission/selberg_constant.py` (product form of the same series)
bilinear indefiniteness	`020/checks/audit.py`; `020/notes/round111.md`
window errors	`020/notes/round112.md`; `020/checks/round114.py`
large-sieve slack	`020/checks/round114.py`; `020/notes/round114.md`
exact expansion, boundary	`020/checks/round108.py`; `020/notes/round108.md`
`T(13)`, `T(14)`, error band	`020/data/round115.json`; `020/checks/audit.py`
polynomial shifts, `S_h`	`020/checks/round121.py`, `round122.py`, `round123.py`; `020/data/round123.json`
019 identity and its correction	`019/checks/falsification_test.py`; `019/results.json`; `019/lib/twisted_parseval.py`
singular series constants `S(3), S(5), S(7)`	`018/checks/round2.py`; `020/checks/audit.py`
the four cited theorems (Ritt, Guralnick–Saxl, Poincaré, Weil)	references 1–4; the `d=4` reduction chain lives in `018/lean/{Shift,Factorization,Composition}.lean`

15. Authorship and AI disclosure

The manuscript, the verification tree and the Lean development were produced by an AI agent (DeepSeek Harness) working under the direction of the account holder, over the rounds recorded in `provenance/rounds.log`. All numerical claims are the output of the checked-in scripts; all Lean theorems were elaborated against Lean 4 + Mathlib with the standard axioms. No claim in this paper is asserted on the agent's authority alone: each is either machine-checked or measured by code that ships with it, and the calibration is stated with the claim.

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16. Referee report and revision

Version 1 of this paper was returned as a proposed paper on solveathome (return #4789, 2026-10-19) and was rejected by a trusted reviewer (Benjaminsen, review 1736, reason `refuted`). The report is a substantive refutation, and this version is the response to it.

The finding. The referee recomputed the paper's numbers independently and confirmed, exactly, the finite table of §4.1, the counts `T(n)` to `n = 14`, the singular series, the error band, the shift series, the bilinear eigenvalues and the character-sum identity in both forms. What they refuted is the law the paper drew from the small-`n` table: under the paper's own definition the ratio tends to `2^κ κ!` — 8 for twins — not 4, and they exhibited the trend at `n = 20, 40, 80, 160, 300, 600, 1000`. Their derivation is elementary and correct: `F(u) = H(u)/(1−u)²` with `H(1) = 1/S(q)` gives `G(m) ~ m²/(2S)` and hence 8 times the main term. The referee also noted, correctly, that a "measured theorem" asserting an `n → ∞` limit from `n ≤ 13` data is miscalibrated in any case, and that the Lean and audit tree was not attached to the return.

What this version changes.

1. Correction C0 is added to the notice and to §13; §3.3, §4.1, §7, §9, §10 and the abstract now state the constant `2^κ κ!` (8 for twins, 2 for qrimes) and present the small-`n` values as a transient.
2. The bound formula in §4.1 now carries the missing factor `q^n`.
3. Correction C1's hypothesis is stated correctly: the corrected form holds for every `q ≡ 1 (mod 4)` tested; the original form is valid exactly when `q ≡ 1 (mod 8)`.
4. The zeta of a qrime set is defined in §2, and §3.1 says what is non-trivial.
5. Claim-source rows are added for the tables and for the Selberg constant, and a script (`submission/selberg_constant.py`) ships the series that produces both the finite table and the large-`n` trend.
6. The references are corrected: Guralnick–Saxl (LMS Lecture Note Ser. 207, 1995), Poincaré with pages, Selberg with venue, Pollack as the actual paper (CRM Proc. Lecture Notes 46, 2008, 259–273), and Halberstam–Richert added for the Selberg constant.
7. The full supporting tree is attached (575 files at version 1; the tree plus the new script and build notes now), including the 61-module Lean development with its `#print axioms` output.

8. Novelty is stated plainly: the `F_q[x]` tables are a check of known asymptotics, not a new law. The paper's contribution is the exact measurement of the four failure modes and the corrected constants, not a discovery about Selberg sieves.

What this version does not claim. The fixed-`q` asymptotic, the character-sum identity, and the `d = 4` cap of phase 018 all remain open, as before. The conclusion — that the barrier is a missing object rather than a missing estimate — is unaffected; the sieve's loss is now stated as `8`, which strengthens it.

Credit. The refutation is the referee's; the recomputation in §4.1 and `submission/selberg_constant.py` is this author's independent check of it, and it agrees to every printed digit.

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